

When Outlier Supervision Fails: Self-Supervised Robustness in Deep Structure-from-Motion

Anonymous CVPR submission

Paper ID *****

Abstract

001 To reconstruct cameras and 3D structure from real im-
002 age collections, learning-based Structure-from-Motion
003 must reject outliers: wrong correspondences that sur-
004 vive geometric verification. The standard recipe trains
005 an outlier classifier with labels derived from an already-
006 trusted reconstruction of the training scenes — su-
007 pervision that presupposes exactly the capability it
008 is meant to provide. We ask when this recipe fails,
009 and show its failures are caused by the labels them-
010 selves, not by domain shift: labels cannot be manufac-
011 tured where robustness is needed most, and retrain-
012 ing the state-of-the-art supervised method (RESfM)
013 on a 60%-contaminated domain with best-case labels
014 still loses $2\times$ to a label-free alternative. That alterna-
015 tive is SS-RESfM: RESfM’s outlier head trained with a
016 self-supervised loss — confident pseudo-labels formed
017 from percentiles of the model’s own reprojection error
018 — with no ground-truth or COLMAP labels anywhere
019 in training or test. Under a strictly controlled protocol
020 (identical tracks, robust bundle adjustment, and train-
021 ing schedule, varying only the outlier mechanism) we
022 compare label-free statistical removal (madweight) and
023 soft reweighting against a from-scratch RESfM base-
024 line across six datasets spanning 0.5%–61% measured
025 outliers. The optimal mechanism flips with contami-
026 nation. Removal wins the high-contamination regime:
027 at 61% plain removal beats the supervised baseline on
028 every seed pair (12.4 ± 2.1 vs. 19.2 ± 3.4) and both
029 beat COLMAP, while at 43% it beats every super-
030 vised seed’s median (1.4 – 1.9 vs. 2.4 – 4.3). Reweighting
031 wins the low-contamination regime, beating the super-
032 vised baseline 2.5 – $2.9\times$ on two of the three clean out-of-
033 distribution splits (BlendedMVS and Olsson; Strecha
034 goes to supervision on every seed) — and, unlike any
035 supervised recipe we test, does so reliably: the label-
036 free arms land in a narrow band on every training
037 seed, whereas the supervised baseline collapses on the
038 clean benchmark in most seeds under every learn-

ing rate, budget, and recipe variant we try (on Ols-
son — 39 essentially outlier-free scenes, 0.5% — ev-
ery MegaDepth-trained supervised variant lands at a
translation error of 7.4–9.1, versus 2.9–3.6 for every
label-free arm). Supervision retains the in-distribution
regime and the cleanest OOD mean (self-supervised
arms within $\sim 38\%$ in distribution). Every compari-
son is backed by five-training-seed bands, a paired-seed
evaluation control, and an environment-noise study —
in a benchmark family where prior work fixes a single
seed. We trace this to a structural circularity: out-
lier labels require a trusted reconstruction of the very
scenes whose contamination makes reconstruction hard
(for any gold-standard labeler, not just COLMAP),
and we measure the resulting label ceiling: regenerat-
ing supervision from scratch, label recall collapses from
0.82 to ~ 0.3 above $\sim 15\%$ contamination, precisely
where robustness is needed. Label-free adaptivity is
thus a missing ingredient for robust deep SfM.

1. Introduction

Structure-from-Motion (SfM) recovers camera poses
and 3D structure from unordered image collections.
Its correspondences are heavily contaminated by out-
liers (wrong matches that survive pairwise geometric
verification), particularly in Internet photo collections
where repeated and symmetric structure produces spu-
rious epipolar geometries. Classical pipelines reject
outliers with RANSAC and robust bundle adjustment
(BA); recent deep equivariant methods (ESfM [2],
RESfM [1]) instead learn a sets-of-sets permutation-
equivariant network that regresses cameras from point
tracks and, in RESfM, adds a supervised per-point in-
lier/outlier classification module. RESfM’s labels are
derived from a COLMAP [4] reconstruction in two
stages: first by track membership, then refined by a
4-pixel reprojection test against COLMAP-pose trian-
gulation [1].

076 We take a different route. Rather than supervise
077 the head with COLMAP labels, SS-RESfM trains it
078 self-supervised: an adaptive confidence-weighted loss
079 forms pseudo-labels from percentiles of the model’s own
080 reprojection error, so no ground-truth or COLMAP
081 outlier labels are used anywhere. On top of it we
082 study a question RESfM’s single supervised mechanism
083 leaves open: is one outlier-handling mechanism
084 optimal across the contamination levels SfM encounters? It is not. Varying only the mechanism, the best
085 one flips with contamination — hard removal high, soft
086 reweighting low, with the supervised classifier retaining
087 the typical in-distribution scene and the cleanest OOD
088 mean (Fig. 1). Plain removal beats the supervised
089 baseline on every seed at both 43% and $\sim 60\%$, while
090 madweight (MAD-based removal with soft-weighted
091 survivors, no learned threshold) matches it there and
092 beats COLMAP. The reason is structural: outlier labels
093 must come from a trusted reconstruction of the
094 very scenes whose contamination makes reconstruction
095 hard, a circularity binding any gold-standard labeler.
096 A self-supervised head consumes only the tracks, so
097 training data becomes as unlimited as unlabeled imagery.
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100 Contributions. (1) The label ceiling, measured and
101 shown causal: outlier supervision requires a trusted reconstruction of the very scenes whose contamination
102 makes reconstruction hard. We measure the resulting label collapse (recall $0.82 \rightarrow \sim 0.3$ above $\sim 15\%$ contamination) and prove it is causal, not domain shift:
103 retrained on a 60%-contaminated domain with best-case labels, supervision still loses $2\times$ to label-free removal (§5.5). (2) The complementarity finding: under
104 an identical tracks+BA+schedule pipeline isolating the mechanism alone, the optimal outlier mechanism flips
105 with contamination — removal at high contamination (beating the supervised baseline on every seed pair at
106 $\sim 60\%$), soft reweighting on clean OOD data ($2.5\text{--}2.9\times$, seed-reliably) — with supervision retaining a curated middle band. (3) SS-RESfM, the instrument of
107 the study and a constructive answer: RESfM’s outlier handling made entirely self-supervised via confident
108 pseudo-labels from the model’s own reprojection-error percentiles — no ground-truth or COLMAP labels anywhere — matching the supervised baseline where labels
109 are sound and surpassing it where they are not; plus madweight, which needs not even the head, removing by robust reprojection statistics (MAD) and
110 beating COLMAP where COLMAP’s own labels cannot be manufactured. (4) A seed- and environment-level reliability methodology for post-BA SfM benchmarks:
111 five-training-seed bands for every cell, a paired-seed evaluation control (2,352 evaluations showing the

129 evaluation seed is immaterial), an environment-noise
130 floor for post-BA means, and per-seed-pairing claim
131 verification — practices absent from this benchmark
132 family, where prior work fixes a single seed, and which
133 surface findings (bimodal clean-OD failures, basin-dependent escapes) that mean $\pm$ std alone would hide.
134 (5) RUC-SfM, a paradigm-agnostic robustness-under-contamination benchmark: standardized track suites
135 for five public OOD datasets plus matched training
136 pools spanning a measured 0.5–61% contamination
137 axis; a pose-level evaluation contract open to any SfM
138 paradigm (tracks, images, view graphs, or pointmaps
139 in — cameras out); a five-seed protocol reporting both
140 metric families plus a catastrophic-cell census; and a
141 machine-verified nine-method reference grid — released
142 with builders, evaluator, and configurations.
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2. Related Work 145

146 Deep equivariant SfM. Learning-based SfM re-
147 gresses cameras from point tracks with permutation-
148 equivariant networks: ESfM [2] casts uncalibrated
149 SfM as deep projective factorization trained with an
150 unsupervised reprojection loss, GASfM [3] a graph-
151 attention variant; both assume outliers were already
152 removed by two-view verification. RESfM [1] targets
153 contaminated internet collections with a sets-of-
154 sets architecture plus a supervised per-point classification
155 module and robust BA, reporting accuracy near
156 its clean-track upper bound (“ESfM*”) and on par
157 with Theia/GLOMAP. End-to-end pointmap pipelines
158 (VGGSfM, MAST3R, DUST3R) target far smaller image
159 sets. Concurrently, VGPA [10] moved robustness
160 to the pose-averaging stage with a self-supervised consistency
161 objective — independently corroborating that
162 label-free robustness is the frontier, but at a different
163 stage and input modality, so direct comparison is confounded.
164 We propose no new architecture: we reuse
165 the RESfM/ESfM backbone and isolate the outlier-
166 handling stage and its supervision.

167 Outlier rejection, robust BA, and TTT. Outliers that
168 survive RANSAC are the central difficulty in internet-
169 scale SfM; classical pipelines suppress them with M-
170 estimators and robust BA, RESfM casts rejection as
171 supervised classification. We contrast that against
172 label-free removal via the median absolute deviation
173 (MAD) [5] and soft reweighting, whose relative merit
174 is contamination-dependent. RESfM’s robust BA is
175 applied identically across all arms, and per-scene test-
176 time fine-tuning is an orthogonal option ($\pm$ TTT) applied
177 uniformly.

3. Method

3.1. Background

The input is a (sparse) point-track tensor M of m tracks across n cameras, built by concatenating pairwise RANSAC-verified SIFT matches. The sets-of-sets permutation-equivariant network [1, 2] maps M to per-camera poses and a per-point outlier score $s \in [0, 1]$ (points in the same track may receive different scores). Crucially, unlike RESfM, whose classification head is trained with a BCE loss against COLMAP-derived inlier/outlier labels, SS-RESfM trains this head self-supervised, with an adaptive confidence-weighted loss: per forward pass it forms confident pseudo-labels from percentiles of the model's own reprojection error (lowest $p\%$ inliers, highest $p\%$ outliers) and applies BCE only to those. No ground-truth or COLMAP outlier labels are used at any point. A reprojection error e is available for every observation once an initial reconstruction exists. All mechanisms below consume s (and, for madweight, e), produce a per-observation weight w (or a hard keep/drop decision), and feed the result to the identical robust-BA stage (Fig. 2).

3.2. Outlier-handling mechanisms

We compare four mechanisms, illustrated as weight-vs-signal transfer functions in Fig. 2:

- remove (the RESfM analogue): drop a point if $s > \tau$; keep with $w=1$ otherwise. We use RESfM's default $\tau=0.6$ and its $\tau=0.8$ for the low-outlier, single-camera Strecha/BlendedMVS sets [1].
- weight: soft, $w = 1 - s$, no removal.
- hybrid: three-band: keep (s below a low percentile), soft-weight (middle), drop (s above a high percentile).
- madweight (ours): removal is driven by robust reprojection statistics, not the head. An observation is removed if its reprojection error exceeds $\text{med}(e) + \alpha \cdot \text{MAD}(e)$ ($\alpha=2$); surviving observations are soft-weighted by $w = 1 - s$. madweight is the only mechanism that consumes the reprojection branch e and requires no learned threshold for removal.

Each mechanism is evaluated with and without test-time fine-tuning ($\pm\text{TTT}$), in which the head continues to adapt on the test scene.

3.3. Test-time fine-tuning and robust BA

We load a multi-scene checkpoint, run a per-scene forward pass to obtain s and e , apply the mechanism, fine-tune per scene (~ 1000 iterations) under the reprojection objective, and apply robust BA. This protocol, the seed, and the schedule are held fixed across all

Table 1. Datasets: outlier rates (RESfM-reported vs. measured on our tracks) and regime.

Dataset	RESfM% ours%		Regime
Strecha [8]	2.7	1.7	clean, LIDAR GT
BlendedMVS [9]	3.6	3.1	low-outlier
MegaDepth [7]	25.6	25.4	contam., in-dist.
1DSfM [6]	31.9	43.3	contam., OOD
1DSfM-hard	–	~ 60	extreme, OOD
Olsson [11]	–	0.5	outlier-free

arms; only the mechanism block changes.

4. Experiments

4.1. Datasets and contamination

We evaluate on six datasets spanning a wide contamination range. Alongside the RESfM paper's quoted outlier fractions we report the contamination measured on our own tracks (robust ground-truth-pose triangulation with the standard 4px cutoff, applied uniformly to every dataset); see Table 1. The suites, splits, and labels are released as RUC-SfM, a standing benchmark defined at the scene + contamination + protocol level, not the input-format level: any paradigm may consume tracks, images, view graphs, or pointmaps and submit per-scene poses with a declared registered set, five times from independent training runs; the evaluator reports both metric families, seed bands, and the catastrophic-cell census (supplementary), with the paper's nine-method five-seed grid as the initial, machine-verified leaderboard. On MegaDepth (tracks identical to RESfM's) the measurements agree (25.4 vs 25.6), validating the protocol; our 1DSfM rebuild (43.3%) is markedly harder than the tracks behind RESfM's quoted 31.9%. This yields a falsifiable prediction, stated in advance: since the winning mechanism is a function of contamination (§5.5), on the authors' $\sim 32\%$ tracks we expect supervision to be competitive, with the label-free advantage emerging toward 43% and dominating by 60%. 1DSfM-hard = the five scenes unused by the standard split, same pipeline, 300 cameras each. Networks train on 27 MegaDepth scenes; all other datasets are OOD.

4.2. Protocol, baselines, and track provenance

Following RESfM's public code and default configuration, all networks train for 20,000 epochs (batch size, image subsampling, the eleven-milestone decay schedule, and validation-based early stopping identical across arms), with five training seeds (20–24) per cell. Two deliberate settings are disclosed. First, the base learning rate is per-method: 10^{-3} for the supervised arms (RESfM's released value), 10^{-4} for

the self-supervised arms, whose pseudo-label loss converges early (best checkpoint $\sim 7.5k$ vs. $\sim 16.5k$ epochs); neither is truncation-limited. Second, the infinity-point margin and the checkpoint-selection metric are held identical across arms rather than at RESfM's released values — a deviation that demonstrably favors the baseline: under the authors' own Accuracy-based selection it is weaker on every dataset (RESfM-faithful row in Table 2; MegaDepth 0.50 vs. 0.34), so all supervised-baseline margins are lower bounds, and its clean-benchmark failures persist under its own rule (Olsson 8.8, BlendedMVS rotation 53°). We report shallow (1×3) and deep (2×3) architectures.

Fair-comparison anchor. Our controlled baseline is RESfM-from-scratch: trained by us with the identical setup as our arms (same 27 scenes, sampling, epochs, seed), so only the outlier mechanism differs. We additionally report RESfM-released (authors' weights) and RESfM (paper).¹

Track provenance. MegaDepth tracks and the 36 test scenes are identical to RESfM (directly comparable to published numbers). The OOD tracks are built by us from the released images and ground truth (RESfM did not release OOD tracks), so OOD comparisons are controlled within our pipeline and not directly comparable to RESfM's published OOD numbers. RESfM's released weights yield 0.209 on MegaDepth in our harness (vs. reported 0.169; unchanged, 0.217, under the authors' exact pinned software stack), isolating the residual gap to solver-level nondeterminism in robust BA (supplementary); the stack-replication study there shows the no-TTT arms are insensitive to environment, while TTT means inherit solver-level noise.

A method-isolated comparison, not a SOTA claim. RESfM-from-scratch shares every pipeline component with our arms and differs only in the outlier mechanism — RESfM's own protocol for re-training ESfM/GASfM. It is also authors-faithful: five re-trains under the authors' validation metric reproduce its band on every dataset (supplementary). We claim no SOTA: the released checkpoint (0.209) beats either from-scratch arm, but that gap stems from the weights themselves (the authors' code on our tracks reproduces 0.34–0.39) and handicaps both arms equally; RESfM-released and RESfM (paper) are reference rows throughout.

Metrics. Following RESfM we draw conclusions from post-BA mean errors (translation emphasized; rotation in degrees; translation is the aligned camera-position

error in scene coordinates), with medians as supporting statistics — except on 1DSfM, whose mean is dominated by two symmetric-structure failure scenes (Ellis Island, Tower of London) that fail for all methods; there the median is representative. Rotation and N_r for all arms are tabulated in the supplementary; rotation broadly agrees with translation, and under high contamination the label-free arms register more cameras than the supervised baseline (38–43% vs. 37% on 1DSfM-hard). Two registration notions must be distinguished: every learned arm outputs a pose for every camera (pose coverage; COLMAP genuinely fails on 36% of 1DSfM-hard), while N_r counts the cameras surviving robust BA and a 5-pixel quality mask, the set over which errors are computed. Under this stricter notion the plain-removal arms attain their low errors over a reduced set (59%/28% vs. the baseline's 77%/37% on 1DSfM/1DSfM-hard), consistent with aggressive pruning disconnecting weakly-supported cameras; madweight is registration-matched to the baseline (74%/38%), so its comparisons are not coverage artifacts.

5. Results

5.1. Main comparison

Table 2 is the controlled head-to-head under the RESfM-aligned schedule: every row shares tracks, architecture, schedule, robust BA, and seed, so any difference is attributable to the outlier mechanism and its supervision alone. Reading along the contamination axis: (1) Clean OOD (Olsson 0.5%, Strecha 1.7%, BlendedMVS 3.1%): the label-free arms win seed-robustly where the supervised head misfires: every arm beats the full baseline band on BlendedMVS (0.14–0.24 vs. 0.35 ± 0.04 ; soft arms $2.5\times$ better, and the baseline's 31.7° rotation failure persists on every seed), and by $\sim 3\times$ on Olsson against four of five baseline seeds (3.0 ± 0.3 vs. 8.2–8.9; one seed escapes at 2.69 — the collapse is basin-dependent, frequency 4/5). The deficit is tail-concentrated: each collapse seed fails a single scene catastrophically (dtu106, error ~ 240 ; the escaping seed lacks exactly this failure), ex-dtu106 the baseline's means (2.2–2.6) match madweight's (2.6–3.3) and pooled medians tie — but the adaptive label-free arms never produce such an outlier scene on any seed, while weight+TTT, sharing the unrescued trunk, fails it 5/5 (elsewhere best-in-class, ex-dtu106 1.7–2.3). The failure is unpredictable scene-level catastrophic basin selection — the tail behaviour that seed-level reliability analysis exists to expose, and that dataset means alone would misattribute. Figure 3 shows the moderate form: on an essentially outlier-free scene the super-

¹The ESfM row likewise follows the unified 20k-epoch protocol. ESfM's own released multi-scene (calibrated) recipe trains for 30k epochs plus a 500-epoch per-scene fine-tune; we deliberately hold the schedule fixed across all rows so that the outlier mechanism remains the only controlled variable.

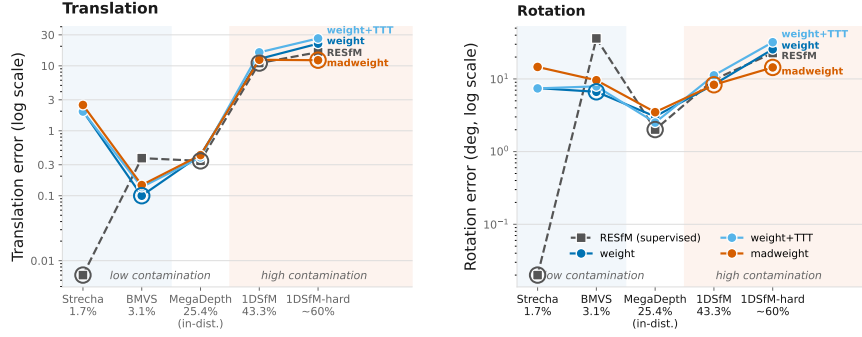


Figure 1. The optimal outlier mechanism flips with contamination. Translation (left) and rotation (right) means (log scale) of the self-supervised arms vs. supervised RESfM, five of the six datasets ordered by measured contamination (Table 2 data). Circled = best per dataset. The rotation panel mirrors the translation ordering, with a self-supervised arm best on 1DSfM and the supervised BlendedMVS failure even more pronounced (36°).

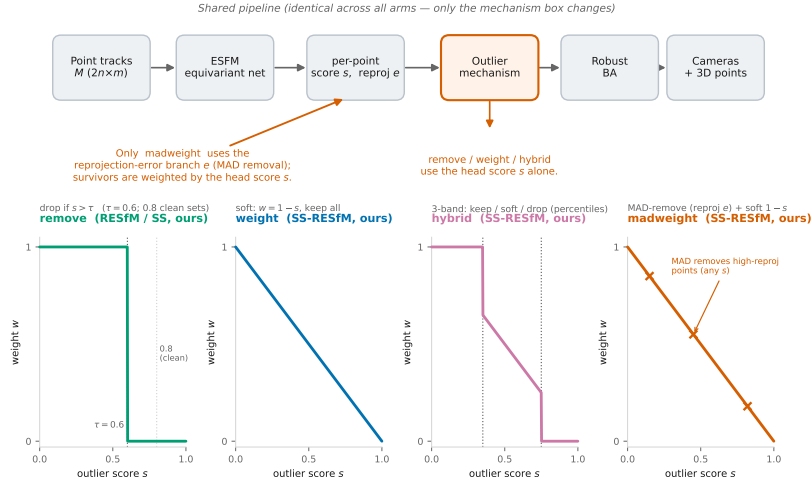


Figure 2. Method. A shared pipeline (top) in which only the outlier-mechanism block changes. The four mechanisms as weight-vs-signal transfer functions (bottom). Only madweight consumes the reprojection-error branch e , and only for its removal step; its surviving points are still weighted by the head score s ($w=1-s$). The other three mechanisms use s alone.

vised reconstruction degenerates to a compressed cluster (16.5° rotation), while the label-free arm reproduces the ground-truth arc nearly exactly. Nor is the collapse a from-scratch artifact: the released checkpoint also collapses on Olsson (9.10; 9.18 under the authors' pinned stack), as does the authors-faithful configuration (8.77). Strecha is the exception: RESfM's mean wins on every seed (0.006–1.67 vs. ~ 2), though the best label-free median ties it (0.005) at seed 20. (2) In distribution (MegaDepth), supervised RESfM retains mean and median. Its mean is strongly seed-dependent (0.37 ± 0.12 over five seeds; its best, 0.19, approaches the released checkpoint) while its median is stable (0.07 ± 0.02); the deterministic self-supervised arms band at 0.51 ± 0.08 and the TTT arm at 0.40 ± 0.11 over five seeds; a $\sim 38\%$ band-level gap without labels, not a

win (against the authors-faithful configuration, 0.496, the arms reach parity, and the TTT arm is ahead). Medians tell the same story more tightly (0.13–0.20 vs. 0.07). (3) Moderate contamination (1DSfM, 43%) has no stable winner between madweight (11.0 ± 1.7) and the baseline (10.5 ± 1.5 ; overlapping bands), so we decline a directional claim for madweight. The removal family leads on both statistics in band terms: remove+TTT 8.9 ± 1.2 vs. the baseline's 10.5 ± 1.5 (separated means, overlapping tails), and plain remove's medians (1.7 ± 0.2) beat every baseline seed pairwise at five seeds each (1.4–1.9 vs. 2.4–4.3; its means, 9.1 ± 2.2 , remain failure-scene-dominated): a label-free win at 43% (the supplementary matrix table), established against the baseline's own five-seed band. (4) Extreme contamination (1DSfM-hard, $\sim 60\%$): removal beats weighting

robustly, madweight (18.9 ± 4.1) statistically matches the baseline (19.2 ± 3.4), and plain remove goes further: 12.4 ± 2.1 , below the baseline on every one of the 25 seed pairs (worst seed 15.8 vs. the baseline's best 16.2; medians 10.8 ± 2.8 vs. 16.5 ± 4.8), and below the released checkpoint (15.29) — the extreme-contamination win holds against every supervised variant we can construct or obtain, and all label-free arms are far ahead of COLMAP (§5.4). Rotation (Fig. 1, right) follows the same ordering.

Classical baselines. GLOMAP (global) and COLMAP (incremental) on the same tracks with known intrinsics sharpen the classical/deep trade-off. GLOMAP is far the best method on pristine calibrated data (Strecha 0.047, an order of magnitude below every learned method) and registers 96–100% of cameras, but under contamination its translations collapse (3.3 on MegaDepth, 27.2 at 43%, 35.4 at 60%, vs. 9.1 and 12.4 for label-free removal) while its rotations stay strong (8.9° at 43%, beating the supervised baseline's 9.6°): a positioning failure, not a loss of orientation structure. The comparison is not coverage-matched — GLOMAP keeps 96% of cameras at 43% where plain removal keeps 58%.

5.2. The complementarity finding

Fig. 1 shows Table 2 as a picture. Between supervision regimes, the self-supervised wins concentrate where a label-free design should help: on clean OOD data, where a classifier trained on contaminated MegaDepth misfires (RESfM's BlendedMVS rotation error explodes to 36°), and at extreme contamination, where supervision can no longer be manufactured (§5.3). Within the self-supervised family the ordering is monotone in contamination: weighting best at 2–3%, a tie in distribution, removal dominant from 43% upward (12.4 ± 2.1 vs. ~ 23 at 60%). This mechanism-level flip — our central question's answer — is stable across schedules and seeds.

Learning-rate robustness, the full mechanism matrix, the MAD/STD/Huber ablation, per-scene selection, and qualitative results: supplementary.

5.3. Why label-free removal wins under high contamination

Two factors erode the supervised advantage. First, domain shift: a decision boundary fit to MegaDepth does not transfer to OOD outlier statistics — and even in distribution the classification is mediocre by RESfM's own ablation (recall 72.1% at precision 44.1%, their Table 7), with their Appendix D crediting much of the accuracy to robust BA. Second, the supervision itself degrades: RESfM's labeling assumes a COLMAP re-

construction and GT poses per training scene, available for curated data and lacking exactly where robustness is needed. madweight needs neither, recomputing its outlier test per scene from reprojection statistics.

Measuring the label ceiling. On 18 scenes spanning the axis we regenerate labels as one must for a new dataset (from-scratch COLMAP, same tracks, known intrinsics, no GT poses) and score them against GT-pose reference labels (coverage = fraction of observations receiving any verdict); Table 3. Precision stays high, but recall collapses from 0.82 to 0.26–0.39 and coverage halves above $\sim 15\%$ contamination: the corrupted reconstruction absorbs enough outliers that most reproject acceptably, and half the observations get no label. A head trained on these labels is taught that most true outliers are inliers, precisely where robustness matters. Below $\sim 15\%$ the labels are excellent — delimiting where the supervised recipe is sound.

5.4. Extreme contamination: the 1DSfM-hard set

To probe beyond 1DSfM's 43%, we evaluate on the five held-out 1DSfM-hard scenes ($\sim 60\%$ measured contamination, 300 cameras each), adding COLMAP (incremental mapping on the same tracks, known intrinsics) and the full mechanism grid (supplementary table). The ordering extrapolates: over seeds 20–24, plain remove attains 12.4 ± 2.1 translation, below the supervised band (19.2 ± 3.4) on every seed pair (medians 10.8 vs. 16.5) — on the field's error-over-registered convention; priced for coverage the ordering inverts (removal registers 29% and is last on RA-AUC while winning 80% of cells; both lenses in the supplementary); madweight (18.9 ± 4.1) matches the baseline; every other mechanism is at 16.9 or worse at seed 20; and COLMAP loses (20.9) even though its error covers only the 64% of cameras it registers. Errors are large for everyone here, but the trend matches the label-ceiling prediction: the supervised advantage vanishes exactly where supervision can no longer be manufactured. Strikingly, plain remove on contaminated tracks also beats ESfM* on oracle-cleaned tracks (12.4 ± 2.1 vs. 18.12 ± 4.35 ; 23 of 25 seed pairs): deleting 60% of observations — even correctly — starves tracks below the multi-view minimum, so perfect labels would not rescue removal-by-oracle. How labels are used matters as much as having them.

5.5. Training on the contaminated domains: the label ceiling is causal

The label-ceiling analysis predicts that supervision fails at high contamination because the labels themselves become untrustworthy — not because of domain shift.

Table 2. Main controlled comparison (RESfM-aligned schedule). Mean / median post-BA translation error. All rows share tracks, architecture, schedule, robust BA, and seeds; only the outlier mechanism and its supervision differ. Upper block = SS-RESfM arms (five-training-seed bands, mean $\pm$ std of per-seed dataset means), as is the supervised baseline; reference and classical rows are single runs, except § RESfM (released): a band over five evaluation seeds on the authors’ fixed weights (network held fixed, spread ≤ 0.008 — evaluation noise is negligible; training seeds are what matter). Best per column in bold. ESfM* = vanilla ESfM on ground-truth-cleaned tracks (clean-track reference; mean failure-scene-dominated, as in RESfM’s own Table 1).

Arm	MegaDepth (25.4) $N_c=13126$	1DSfM (43.3) 4563	1DSfM-hard (60) 1497	Strecha (1.7) 54	BMVS (3.1) 225	Olsson (0.5) 3858
Translation error (mean / median), the primary metric						
madweight	0.51 $\pm$ 0.08/0.16	11.0 $\pm$ 1.7/4.9	18.9 $\pm$ 4.1/13.4	2.65 $\pm$ 0.25/1.16	0.14 $\pm$ 0.02/0.05	3.0 $\pm$ 0.3 /1.4
weight	0.51 $\pm$ 0.09/0.20	14.4 $\pm$ 3.0/9.6	22.9 $\pm$ 2.8/15.6	1.97 $\pm$ 0.09/0.04	0.14 $\pm$ 0.03 /0.02	2.9 $\pm$ 0.3/1.1
weight+TTT	0.40 $\pm$ 0.11/0.18	15.7 $\pm$ 2.1/10.8	22.2 $\pm$ 3.2/15.3	2.00 $\pm$ 0.05/0.12	0.14 $\pm$ 0.02/ 0.01	8.0 $\pm$ 0.3/0.6
remove	0.51 $\pm$ 0.08/0.13	9.1 $\pm$ 2.2/ 1.7 $\pm$ 0.2	12.4 $\pm$ 2.1 /10.8	3.20 $\pm$ 0.11/2.03	0.24 $\pm$ 0.06/0.17	3.4 $\pm$ 0.6/1.7
remove+TTT	0.56 $\pm$ 0.07/0.14	8.9 $\pm$ 1.2 /2.0	15.2 $\pm$ 1.0/11.3	3.01 $\pm$ 0.07/1.70	0.22 $\pm$ 0.10/0.18	3.6 $\pm$ 0.4/1.9
RESfM-scratch (supervised)	0.37 $\pm$ 0.12 / 0.07	10.5 $\pm$ 1.5/3.4	19.2 $\pm$ 3.4/16.5	0.39 $\pm$ 0.72 / 0.01	0.35 $\pm$ 0.04/0.33	7.4 $\pm$ 2.6/ 0.95
RESfM (released) §	0.203 $\pm$ 0.008	10.71 $\pm$ 0.27	15.29	0.20 $\pm$ 0.00	0.33 $\pm$ 0.00	9.10
RESfM (paper)	0.169	—	—	—	—	—
RESfM-faithful (authors’ ckpt selection)	0.496	9.75	17.58	0.144	0.367	8.77
ESfM* (clean tracks)	0.60 $\pm$ 0.18	13.24 $\pm$ 0.52	18.12 $\pm$ 4.35	1.14 $\pm$ 0.45	3.79 $\pm$ 3.51	9.36 $\pm$ 1.28
ESfM (same tracks, no outlier mech.)	0.71 $\pm$ 0.11	18.42 $\pm$ 0.75	21.76 $\pm$ 0.80	2.03 $\pm$ 0.14	6.45 $\pm$ 3.75	5.91 $\pm$ 3.32
GASfM (released ckpt, our robust BA)	2.25	25.84	36.49	1.14	11.13	4.45
GLOMAP (classical, global)	3.33	27.21	35.42	0.047	0.339	3.17
COLMAP (classical, increm.)	0.62 $\pm$ 0.03	6.29 $\pm$ 0.42	20.4 $\pm$ 1.1	0.03 $\pm$ 0.00	0.01 $\pm$ 0.00	0.20 $\pm$ 0.00

Buddah Tooth Relic Temple (Olsson, 0.5% contamination), both trained on MegaDepth: the clean-OOD collapse

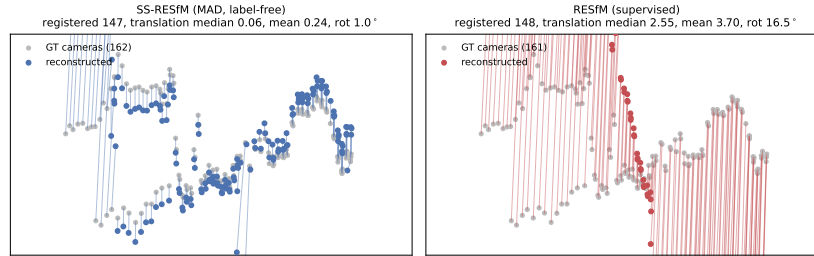


Figure 3. The clean-OOD collapse, qualitatively (Buddah Tooth Relic Temple, Olsson, 0.5% contamination; both networks trained on MegaDepth, seed 20): top-view camera centers in the evaluation frame; whiskers join each reconstructed camera to its ground-truth position. The label-free madweight arm (left) lands on the ground-truth layout (translation median 0.06, rotation 1.0°); the supervised baseline (right) registers the same number of cameras but collapses them toward a line with large systematic displacements (median 2.55, rotation 16.5°) — on a scene with almost no outliers to reject.

Table 3. The label ceiling: quality of COLMAP-regenerated supervision vs. contamination.

Contam.	n	Cover%	Prec.	Recall	COLMAP t -err
0–5%	6	99.4	0.76	0.82	0.004
5–15%	4	89.6	0.90	0.83	0.014
15–30%	2	71.7	0.83	0.26	1.02
30–45%	3	55.7	0.74	0.39	0.82
45–60%	3	55.1	0.87	0.39	11.0

vised labels generated at reference quality, by the same Appendix-C procedure as MegaDepth’s but against each dataset’s real GT poses — the best case supervision could ever obtain). Table 4 reports five-seed bands on the held-out scenes.

Four findings (supplementary). (i) At 60%, supervision loses on its own training domain: the supervised head, trained on these scenes with reference-quality labels, is beaten $\sim 2\times$ at band level by madweight (12.1 $\pm$ 6.1 vs. 23.4 $\pm$ 7.8; four of five madweight seeds beat every supervised seed; medians 9.8 vs. 20.6). “Just train the classifier there” does not work: the high-contamination failure is caused by labeling at contamination, not by domain shift. Sharper still: the

We test this causally by making the OOD domains in-distribution: for each dataset we split its scenes into train/validation/held-out test and retrain both recipes on the domain itself (identical protocol; super-

Table 4. In-distribution comparison on the (formerly OOD) domains, ordered by contamination: mean translation over held-out test scenes, mean $\pm$ std over five training seeds. Supervised labels are GT-derived (best case).

Arm	Olsson 0.5%	BMVS 3.1%	1DSfM 43%	hard 60%
madweight	1.97 $\pm$ 0.94	0.093 $\pm$ 0.099	6.37 $\pm$ 0.20	12.1 $\pm$ 6.1
weight	3.05 $\pm$ 1.39	0.114 $\pm$ 0.006	10.65 $\pm$ 2.49	28.1 $\pm$ 5.9
weight+TTT	1.60 $\pm$ 0.81	0.081 $\pm$ 0.046	9.91 $\pm$ 2.39	29.3 $\pm$ 8.4
remove	2.94 $\pm$ 0.27	0.228 $\pm$ 0.080	6.68 $\pm$ 1.44	19.4 $\pm$ 7.7
RESfM (sup., GT)	2.03 $\pm$ 1.86	0.260 $\pm$ 0.149	4.97 $\pm$ 2.08	23.4 $\pm$ 7.8
ESfM (no mech.)	1.45 $\pm$ 1.44	2.36 $\pm$ 3.27	10.20 $\pm$ 2.97	23.1 $\pm$ 5.9

supervised head with best-case labels performs identically to no outlier mechanism at all at this contamination (vanilla ESfM trained in-domain: 23.14 $\pm$ 5.89 vs. supervised 23.4 $\pm$ 7.8) — at 60%, GT-supervised classification adds nothing over doing nothing, while label-free removal halves both. The vanilla row completes a symmetric statement: GT-label supervision matches no mechanism at all at both extremes of the axis — identical at 60% (23.4 vs. 23.1) and overlapping at 0.5% in-domain (2.03 $\pm$ 1.86 vs. 1.45 $\pm$ 1.44) — earning its keep only in the curated middle band (25–43%). The absolute errors are not uniformly “large for everyone”: madweight produces usable reconstructions where supervision fails outright, and where supervised errors look lower they are computed over 3–4 $\times$ fewer registered cameras (per-scene decomposition and the error-vs-coverage trade-off: supplementary). (ii) The BlendedMVS failure was never domain shift either: supervision trained on BlendedMVS lands at 0.20–0.37 in four of five seeds — indistinguishable from the MegaDepth-trained baseline evaluated OOD (0.35 $\pm$ 0.04) — with a single escape seed (0.02), the same bimodality as its clean-OOD behaviour, while label-free weighting is seed-stable at 0.114 $\pm$ 0.006. Class imbalance alone cannot explain this: at 0.5% (Olsson), where positives are rarer still, supervision trained in-domain reaches 0.41–1.08 in three of five seeds — extreme positive scarcity does not preclude learning — whereas BlendedMVS never reaches that level on any seed; the BlendedMVS failure is a dataset-specific dead zone (its rotation pathology is the leading suspect), not a general low-contamination bound. (iii) In-domain training helps far less than raw means suggest: the apparent improvement from 7.4–9.1 (MegaDepth-trained) to 2.03 $\pm$ 1.86 (in-domain) conflates scene sets, since the in-distribution split excludes dtu106 — the single scene driving the OOD means — by placing it in training. On the same eight test scenes the MegaDepth-trained model scores 2.72 (1.31–3.53) against 2.03 $\pm$ 1.86: overlapping bands, with the in-domain model the more bimodal. The seed-reliability asymmetry sur-

vives: weight+TTT (1.60 $\pm$ 0.81) leads, madweight (1.97 $\pm$ 0.94) matches the supervised mean at half the spread, and no adaptive label-free arm has a failure seed — supervision’s clean-data advantage is a three-in-five-seeds advantage, not a dependable one. (iv) The supervised advantage inverts between 43% and 60% (4.97 $\pm$ 2.08 vs. madweight 6.37 $\pm$ 0.20 at 43%; a 2 $\times$ loss at 60%), and the complementarity ordering replicates in-distribution — the mechanism flip is a property of the contamination level, not the train/test relationship. Read as a curve, best-case-label supervision wins at 25% and narrowly 43%, is seed-bimodal at 0.5% (tied in mean, far wider in spread), and loses at 3.1% and 60%: two distinct failure modes — a dataset-specific dead zone, and the causal label ceiling that no in-domain re-training can cross — with seed-unreliability shadowing the whole axis.

6. Limitations

(1) No single configuration wins all datasets; per-dataset selection (Table 2) requires knowing the contamination level. This studies the phenomenon, not a turnkey method; an unsupervised selector is future work, as is anchoring with label-free pose averaging (cf. Khatib et al. 10). (2) Supervised RESfM retains in-distribution MegaDepth (self-supervised arms come within $\sim$ 38%) and the clean Strecha mean, and RESfM-released is stronger on MegaDepth: we claim no SOTA. (3) madweight matches rather than beats supervision at moderate/extreme contamination; plain remove leads every baseline seed at $\sim$ 60% but over a reduced registered set (§4.2), and the magnitude of the supervised deficit is learning-rate-dependent (supplementary) — the Olsson collapse and the mechanism flip are not. (4) OOD tracks are self-constructed (not comparable to RESfM’s published OOD numbers); (5) two symmetric-structure scenes fail for all methods [1].

7. Conclusion

SS-RESfM makes deep SfM outlier handling entirely self-supervised: within $\sim$ 38% of supervision in distribution, winning two of three clean-OOD splits on every seed, and beating it at high contamination, where supervision can no longer be manufactured. The optimal mechanism is contamination-dependent, so robust deep SfM should treat outlier handling as adaptive and self-supervised.

References

- [1] Khatib, Kasten, Moran, Galun, and Basri. RESfM: Robust Deep Equivariant Structure from Motion. ICLR 2025. 1, 2, 3, 8

- 609 [2] Moran, Koslowsky, Kasten, Maron, Galun, and Basri.
610 Deep Permutation Equivariant Structure from Mo-
611 tion. ICCV 2021. 1, 2, 3
- 612 [3] Brynte, Iglesias, Olsson, and Kahl. Learning
613 Structure-from-Motion with Graph Attention Net-
614 works. CVPR 2024. 2
- 615 [4] Schönberger and Frahm. Structure-from-Motion Re-
616 visited. CVPR 2016. 1
- 617 [5] Rousseeuw and Croux. Alternatives to the Median Ab-
618 solute Deviation. JASA 1993. 2
- 619 [6] Wilson and Snavely. Robust Global Translations with
620 1DSfM. ECCV 2014. 3
- 621 [7] Li and Snavely. MegaDepth: Learning Single-View
622 Depth Prediction from Internet Photos. CVPR 2018.
623 3
- 624 [8] Strecha, von Hansen, Van Gool, Fua, and Thoen-
625 nessen. On Benchmarking Camera Calibration and
626 Multi-View Stereo for High-Resolution Imagery.
627 CVPR 2008. 3
- 628 [9] Yao et al. BlendedMVS: A Large-Scale Dataset
629 for Generalized Multi-View Stereo Networks. CVPR
630 2020. 3
- 631 [10] Khatib, Galun, and Basri. Learning Global Camera
632 Poses from Noisy View-Graphs for Structure from Mo-
633 tion (VGPA). ECCV 2026. arXiv:2609.09491. 2, 8
- 634 [11] Olsson and Enqvist. Stable Structure from Motion for
635 Unordered Image Collections. SCIA 2011. 3